

[Aim: 100/100 in Maths]

आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #3

#ATK²:

PLF2V

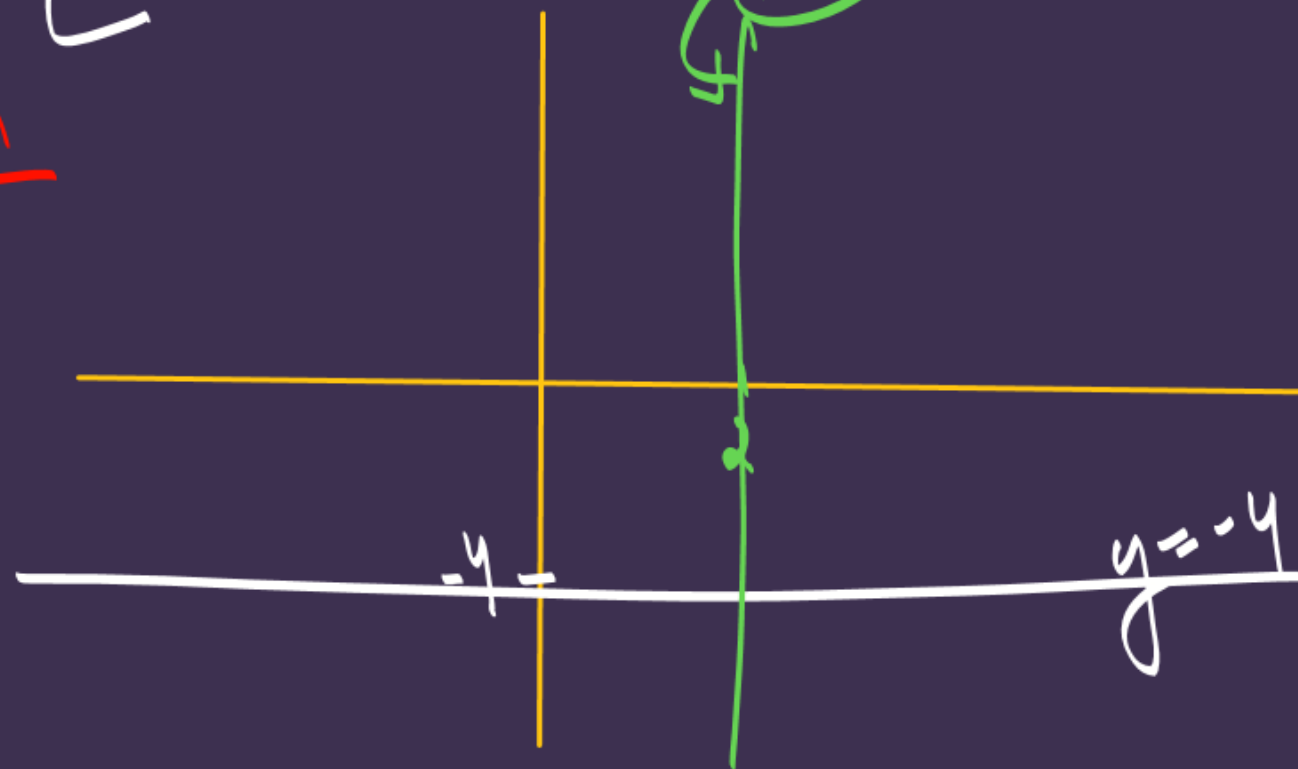
$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}$$

standard form

$$y = -4$$

$x = 2$

$$\left[\begin{aligned} \frac{a_1}{a_2} &\neq \frac{b_1}{b_2} \quad \text{no solution} \\ \frac{a_1}{a_2} &= \frac{b_1}{b_2} = \frac{c_1}{c_2} \quad \text{infinitely many solutions} \\ \frac{a_1}{a_2} &= \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \quad \text{no solution} \end{aligned} \right]$$



#LP: Use graphical method to solve the system of linear equations:

$x = -3$ and $5x - 2y = -5$

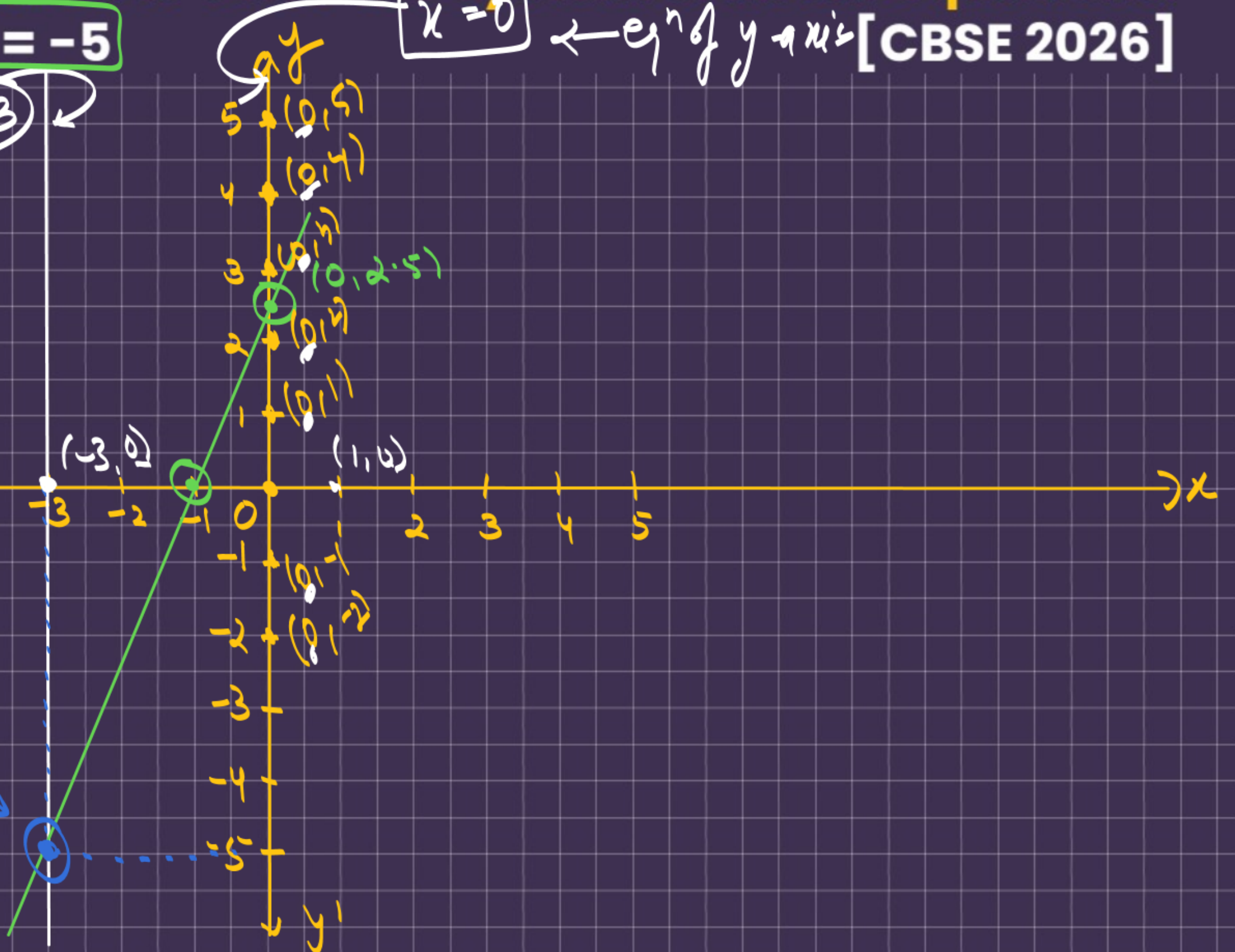
$x = 0$

← eqⁿ of y-axis [CBSE 2026]

$x = -3$

$y = 0$
← eqⁿ of x-axis

Ans $\begin{cases} x = -3 \\ y = -5 \end{cases}$



$$5x - 2y = -5$$

x	0	-1
y	2.5	0

$$(0, 2.5)$$

$$(-1, 0)$$

$$5x - 0 = -5$$

$$x = -1$$

#LP: The point of intersection of the line represented by $3x - y = 3$ and y-axis is given by: [CBSE 2023]

- (A) $(0, -3)$
- (B) $(0, 3)$
- (C) $(2, 0)$
- (D) $(-2, 0)$

$$3x - y = 3$$
$$x = 0$$

put

$$3(0) - y = 3$$
$$y = -3$$

$$(0, -3)$$

find the point of intersection of $3x - y = 2$ and $y = 3$.

put

$$3x - 3 = 2$$

$$3x = 2 + 3$$

$$3x = 5$$

$$x = \frac{5}{3}$$

POI $\rightarrow$

$$\left(\frac{5}{3}, 3 \right)$$

^{Point of Int.}
#LP: The lines k_1 , k_2 and k_3 represent three different equations as shown in the graph below. The solution of the equations represented by the lines k_1 and k_3 is $x = 3$ and $y = 0$ while the solution of the equations represented by the lines k_2 and k_3 is $x = 4$ and $y = 1$. Which of these is the equation of the line k_3 ? [CBSE APQ 2024]

- (A) $x - y = 3$
- (B) $x - y = -3$
- (C) $x + y = 3$
- (D) $x + y = 1$

$(3, 0)$ & $(4, 1)$

lies on k_3

(B) $x - y = -3$

$(3, 0)$
 $3 - 0 = -3$
 $3 \neq -3$

check

(A) $x - y = 3$ ✓

$(3, 0)$ ✓
 $3 - 0 = 3$

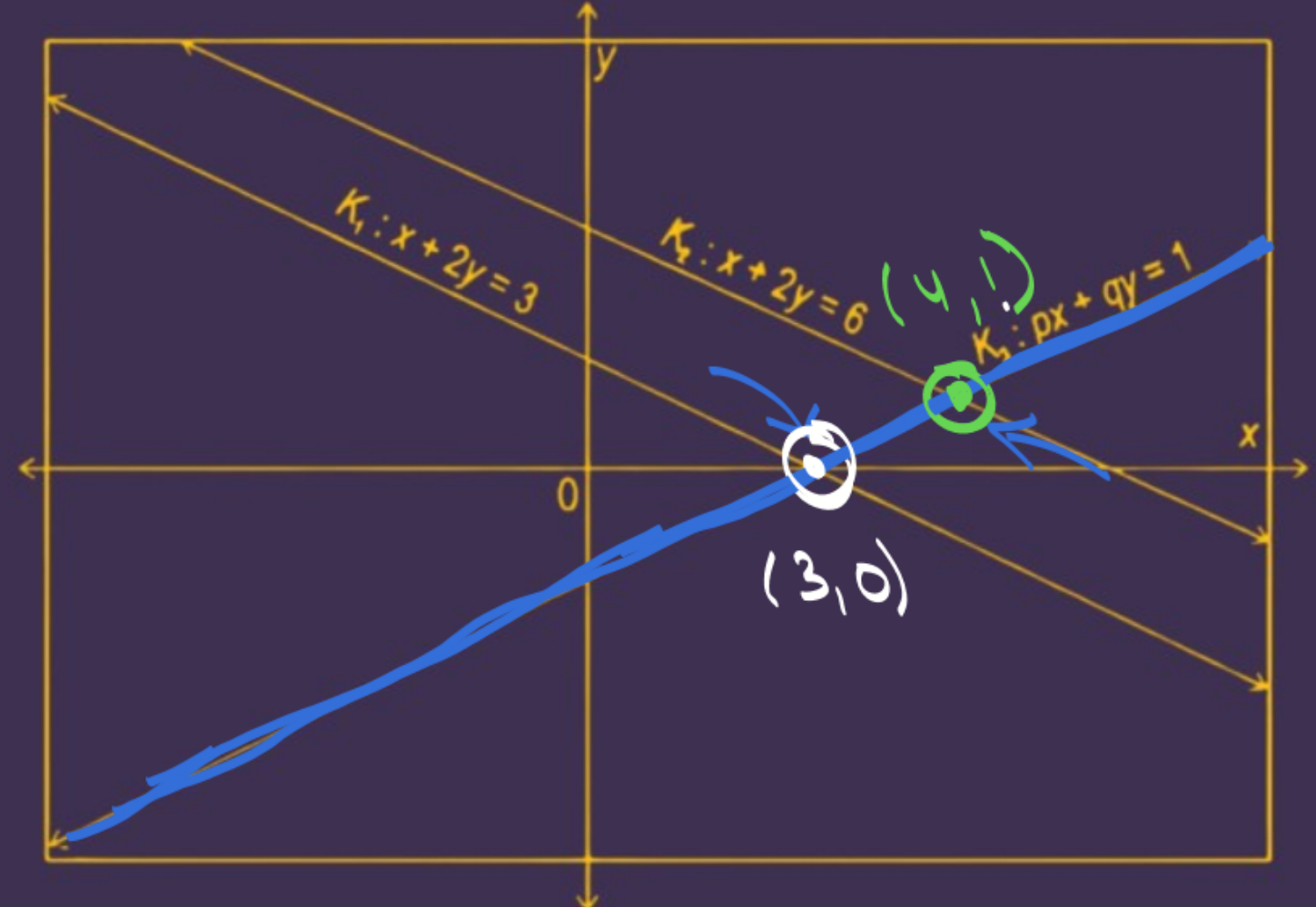
$3 = 3$ ✓

LHS = RHS

$(4, 1)$ ✓
 $4 - 1 = 3$

$3 = 3$ ✓

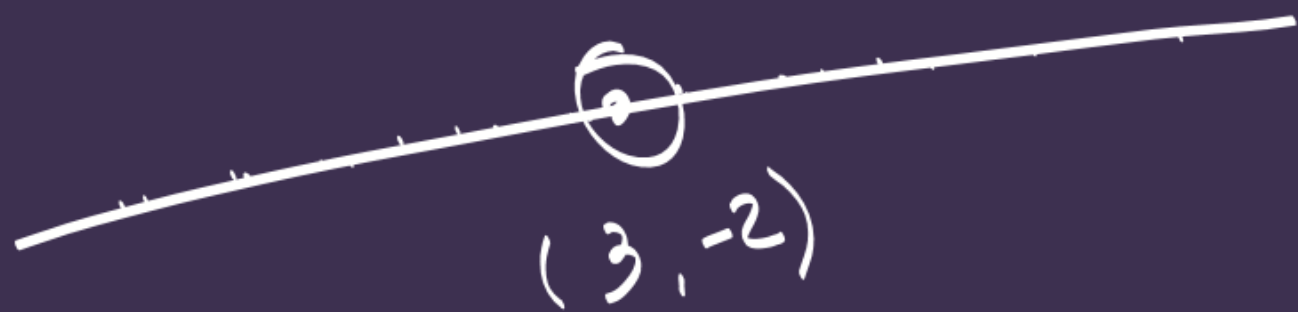
LHS = RHS





all the points on the line
satisfy the eqⁿ of line.

Q: If the point $(3, -2)$ lies on line $3x - 4y = p$. Then find the value of p .



∴ $(3, -2)$ will satisfy the eqⁿ of line

Put $3x - 4y = p$

$$3(3) - 4(-2) = p$$

$$9 + 8 = p$$

$$p = 17$$

Methods of finding solutions: x & y की value

~~Graphical~~

Algebraic Method

substitution

& Elimination

1. Substitution method:

→ एक में value डालकर दूसरे में पक दो!

$$x + 2y = 6 \quad \text{--- (I)}$$

$$2x - 5y = 12 \quad \text{--- (II)}$$

$$x = 6 - 2y$$

$$\Rightarrow 2(6 - 2y) - 5y = 12$$

$$\Rightarrow \cancel{12} - 4y - 5y = \cancel{12}$$

$$\Rightarrow -9y = 0$$

$$y = 0$$

put

$$x = 6 - 2(0)$$

$$x = 6$$

Final Ans $\Rightarrow (6, 0)$

$$\begin{cases} x = 6 \\ y = 0 \end{cases}$$

ii

$$3x - y = 2 \quad \text{--- (I)}$$

$$x + 4y = 3 \quad \text{--- (II)}$$

$$3x - 2 = y \quad \text{--- (III)}$$

put this y in (II)

$$x + 4(3x - 2) = 3$$

$$x + 12x - 8 = 3$$

$$13x = 3 + 8$$

$$13x = 11$$

$$x = \frac{11}{13}$$

put in (III)

$$y = 3x - 2$$

$$y = 3\left(\frac{11}{13}\right) - 2$$

$$y = \frac{33}{13} - 2$$

$$y = \frac{33 - 26}{13}$$

$$y = \frac{7}{13}$$

iii

$$x + y = -2 \quad \text{--- (I)}$$

$$3x + 4y = 3 \quad \text{--- (II)}$$

$$x = -y - 2$$

put in (II)

$$\Rightarrow 3(-y - 2) + 4y = 3$$

$$\Rightarrow -3y - 6 + 4y = 3$$

$$= 6 + 3$$

$$y = 9$$

$$x = -9 - 2$$

$$x = -11$$

$$(-11, 9)$$

#LP: Solve the following system of linear equations:

[CBSE 2024]

$$7x - 2y = 5$$

$$8x + 7y = 15$$

Verify your answer.

#LP: If $2x + y = 13$ and $4x - y = 17$, find the value of $(x - y)$. [CBSE 2024]

$$y = 13 - 2x$$

$$\Rightarrow 4x - (13 - 2x) = 17$$

$$\Rightarrow 4x - 13 + 2x = 17$$

$$\Rightarrow 6x = 13 + 17$$

$$\Rightarrow 6x = 30$$

$$x = 5$$

$$y = 13 - 2(5)$$

$$y = 13 - 10$$

$$y = 3$$

$$5 - 3 = 2$$

#LP: Solve the following using substitution method:

$$\frac{(x+y)}{xy} = 2 \text{ and } \frac{(x-y)}{xy} = 6$$

$$\frac{x}{xy} + \frac{y}{xy} = 2$$

$$\frac{1}{y} + \frac{1}{x} = 2$$

let

$$\Rightarrow \boxed{p+q=2}$$

$$\frac{1}{x} = p$$

$$\frac{x}{xy} - \frac{y}{xy} = 6$$

$$\frac{1}{y} - \frac{1}{x} = 6$$

$$\Rightarrow \boxed{p-q=6}$$

$$\boxed{p+q=2}$$

$$\boxed{p=2-q}$$

$$p=2-(-2)$$

$$p=2+2$$

$$\boxed{p=4}$$

Now

$$\frac{1}{x} = 4 \Rightarrow x = \frac{1}{4}$$

$$\boxed{p-q=6}$$

$$(2-q)-q=6$$

$$2-2q=6$$

$$-2q=6-2$$

$$-2q=4$$

$$\boxed{q=-2}$$

$$\Rightarrow \frac{1}{x} = -2$$

$$\boxed{x = -\frac{1}{2}} \text{ A}$$

#LP: Solve the following :

$$0.5x + 0.2y = 3.4$$

$$\boxed{x - y = 2} \text{---} \textcircled{\text{I}}$$

$$\Rightarrow \frac{0.5x}{10} + \frac{0.2y}{10} = \frac{3.4}{10}$$

$$\Rightarrow \frac{1}{10} (5x + 2y) = \frac{3.4}{10}$$

$$\Rightarrow \boxed{5x + 2y = 34} \text{---} \textcircled{\text{I}}$$

$$\Rightarrow \boxed{x - y = 2} \text{---} \textcircled{\text{II}}$$

HW

Next DPP

THANK YOU COODIESS !!

